Systems Architecture

Karnaugh Maps

Advantages of Karnaugh Maps

Graphical technique for representation and simplification of Boolean expressions

- Graphical Technique
- Handle up to six variables
- Unambiguous
- Two-dimensional
- Simplification *seen* immediately

Karnaugh Map Layout

Adjacent squares (horizontal, vertical, not diagonal) differ by only one variable

	0	1
A	A	$\overline{A}$

	AB	$A\overline{B}$	$\overline{A}\,\overline{B}$	$\overline{A}B$
C	ABC	$A\overline{B}C$	$\overline{A}\,\overline{B}C$	$\overline{A}BC$
$\overline{C}$	$AB\overline{C}$	$A\overline{B}\,\overline{C}$	$\overline{A}\,\overline{B}\,\overline{C}$	$\overline{A}B\overline{C}$

	A	$\overline{A}$
B	AB	$\overline{A}B$
$\overline{B}$	$A\overline{B}$	$\overline{A}\,\overline{B}$

	AB	$A\overline{B}$	$\overline{A}\,\overline{B}$	$\overline{A}B$
CD	ABCD	$A\overline{B}CD$	$\overline{A}\,\overline{B}CD$	$\overline{A}BCD$
$\overline{C}D$	$AB\overline{C}D$	$A\overline{B}\,\overline{C}D$	$\overline{A}\,\overline{B}\,\overline{C}D$	$\overline{A}B\overline{C}D$
$\overline{C}\,\overline{D}$	$AB\overline{C}\,\overline{D}$	$A\overline{B}\,\overline{C}\,\overline{D}$	$\overline{A}\,\overline{B}\,\overline{C}\,\overline{D}$	$\overline{A}B\overline{C}\,\overline{D}$
$C\overline{D}$	$ABC\overline{D}$	$A\overline{B}C\overline{D}$	$\overline{A}\,\overline{B}C\overline{D}$	$\overline{A}BC\overline{D}$

Using Karnaugh Maps

- Adjacent square differ by only *one* variable

  $\overline{A}\,\overline{B}$, $\overline{A}\,B$, $A\,B$, $A\,\overline{B}$ or $A\,B$, $A\,\overline{B}$, $\overline{A}\,\overline{B}$, $\overline{A}\,B$

  $\overline{A}\,\overline{B}\,\overline{C}$, $\overline{A}\,\overline{B}\,C$, $\overline{A}\,B\,C$, $\overline{A}\,B\,\overline{C}$, $A\,B\,\overline{C}$, $A\,B\,C$, $A\,\overline{B}\,C$, $A\,\overline{B}\,\overline{C}$

- Two dimensional view of truth table
- Only plot the 1's
- *torus* or *doughnut* in shape

Views of the Truth Table

A	B	C	D	F	
0	0	0	0	0	
0	0	0	1	0	
0	0	1	0	0	
0	0	1	1	1	$\overline{A}\,\overline{B}\,C\,D$
0	1	0	0	0	
0	1	0	1	1	$\overline{A}\,B\,\overline{C}\,D$
0	1	1	0	0	
0	1	1	1	0	
1	0	0	0	0	
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	1	$A\,\overline{B}\,C\,D$
1	1	0	0	0	
1	1	0	1	1	$A\,B\,\overline{C}\,D$
1	1	1	0	0	
1	1	1	1	0	

$$F = \overline{A}\,\overline{B}\,C\,D + \overline{A}\,B\,\overline{C}\,D + A\,\overline{B}\,C\,D + A\,B\,\overline{C}\,D$$

F	$\overline{A}\,\overline{B}$	$\overline{A}\,B$	$A\,B$	$A\,\overline{B}$
$\overline{C}\,\overline{D}$				
$\overline{C}\,D$				
$C\,D$				
$C\,\overline{D}$				

Karnaugh Map Example 1

F	$\overline{A}\,\overline{B}$	$\overline{A}\,B$	$A\,B$	$A\,\overline{B}$
$\overline{C}\,\overline{D}$				
$\overline{C}\,D$		1	1	
$C\,D$	1			1
$C\,\overline{D}$				

$$F = A\,\overline{B}\,C\,D + \overline{A}\,\overline{B}\,C\,D + A\,B\,\overline{C}\,D + \overline{A}\,B\,\overline{C}\,D$$

$$= (\overline{A} + A)(B\,\overline{C}\,D) + (\overline{A} + A)(\overline{B}\,C\,D)$$

$$= B\,\overline{C}\,D + \overline{B}\,C\,D$$

Karnaugh Map Simplification

1. Plot all 1's
2. Combine adjacent 1's

   Group as many as possible

   (1, 2, 4, 8, 16)
3. OR each grouping

- Products cover multiple squares
- One square may be in more than one group

Karnaugh Map Example 2

F_1	$\overline{A}\,\overline{B}$	$\overline{A}\,B$	$A\,B$	$A\,\overline{B}$
$\overline{C}\,\overline{D}$				
$\overline{C}\,D$				
$C\,D$				
$C\,\overline{D}$				

F_2	$\overline{A}\,\overline{B}$	$\overline{A}\,B$	$A\,B$	$A\,\overline{B}$

$$F_1 = A\,B + \overline{A}\,B\,\overline{C}\,D + B\,C\,D + A\,\overline{B}\,\overline{C}\,\overline{D}$$

$$F_2 = B\,D + \overline{B}\,\overline{C}\,\overline{D} + A\,\overline{B}\,\overline{D} + \overline{A}\,\overline{B}\,C\,\overline{D}$$

Karnaugh Map Example 3

F_3	$\overline{A}\,\overline{B}$	$\overline{A}\,B$	$A\,B$	$A\,\overline{B}$
$\overline{C}\,\overline{D}$				
$\overline{C}\,D$				
$C\,D$				
$C\,\overline{D}$				

F_4	$\overline{A}\,\overline{B}$	$\overline{A}\,B$	$A\,B$	$A\,\overline{B}$

$$F_3 = A\,B + \overline{A}\,B\,\overline{C}\,D + \overline{A}\,B\,C\,D + B\,\overline{C}\,\overline{D}$$

$$F_4 = A\,\overline{C}\,\overline{D} + \overline{A}\,\overline{B}\,C + \overline{C}\,D + A\,\overline{B}\,D$$

Inverted Karnaugh Maps

F_3	$\overline{A}\,\overline{B}$	$\overline{A}\,B$	$A\,B$	$A\,\overline{B}$
$\overline{C}\,\overline{D}$				
$\overline{C}\,D$				
$C\,D$				
$C\,\overline{D}$				

F_4	$\overline{A}\,\overline{B}$	$\overline{A}\,B$	$A\,B$	$A\,\overline{B}$

$$F_3 = A\,B + \overline{A}\,B\,\overline{C}\,D + \overline{A}\,B\,C\,D + B\,\overline{C}\,\overline{D}$$

$$F_4 = A\,\overline{C}\,\overline{D} + \overline{A}\,\overline{B}\,C + \overline{C}\,D + A\,\overline{B}\,D$$

The Don't Care Condition

X = don't care (either 1 or 0)
X = input combination not defined
X = input combination impossible

- Plot X's
- Include X's in groupings

	$\overline{A}\,\overline{B}$	$\overline{A}\,B$	$A\,B$	$A\,\overline{B}$
$\overline{C}\,\overline{D}$		1	X	
$\overline{C}\,D$	1	1	X	
$C\,D$	X	X	X	X
$C\,\overline{D}$		1	X	